

A NEW AGE OF DISCOVERY

The future is almost here

"India's innovation is less than 15 years old"

GLENN SALDANHA MD & CEO, GLENMARK

If you look at the entire drug discovery process for the last century, the focus has been on small molecules and new chemical entities. Chemistry is the backbone for making these substances. Today, globally, there is a dramatic shift taking place into something a lot more novel—biologics. We believe biologics represents a significant opportunity in the next 15-20 years and following that we think therapy will move towards regenerative medicine, stem cell research or personalised medicine. Infectious diseases such as pneumonia and tuberculosis were some of the prime killers in the past. Today, given the way drug therapy has gone, we have eradicated many of them. The challenge for the drug industry today is mainly in the area of chronic diseases, where progression is slow. Clearly, the maximum number of fatalities come from chronic diseases such as heart attack and cancer, primarily predominant because of our high stress lifestyle. We believe diseases will shift from the chronic to the degenerative—where there is a progressive deterioration of some of the tissues of the human body.

HEMANT CHAWLA

Over the last 15 years you have seen a large rush of new drugs and a classic example is Prozac. Even today you have over two million people who experience adverse drug reactions because of small molecules—the fatalities are one lakh. In addition, there are certain new trends in the pharma industry that are paving the way for new therapy. The Human Genome Project is paving the way for new drug treatment and a better understanding of the human immune system. What exactly is a biological drug? It is nothing but a substance made from a living organism or its products and is used for prevention, diagnosis and treatment of diseases. The four main categories are insulin, vaccine, antibodies and therapeutic proteins, of which antibodies are the most promising. Biologics score over chemically-synthesised molecules as they are less toxic.

Within biologics, antibodies are the most exciting area. But what are antibodies? Basically, when the body is under attack from either a bacteria or a virus, also called antigens, the body secretes antibodies to fight the infection and helps destroy them. So this is something which the body synthesises as a natural process. A classic example is chickenpox and smallpox. Those of you who have suffered from these clearly recognise that the likelihood of getting it again is very low. It is purely because the antibodies have been activated by the disease and the body now recognises them and is able to fight subsequent attacks of these diseases. Stem cell is like

INNOVATORS ALL: (From left) Attenborough, session chairperson Avnish Bajaj and Saldanha

